

Synthesis of the leading strand and synthesis of the lagging strand occur concurrently and at the same rate. The lagging strand is so named because its synthesis is delayed slightly relative to synthesis of the leading strand; each new fragment of the lagging strand cannot be started until enough template has been exposed at the replication fork.

Figure 16.18 and **Table 16.1** summarize DNA replication. Please study them carefully before proceeding.

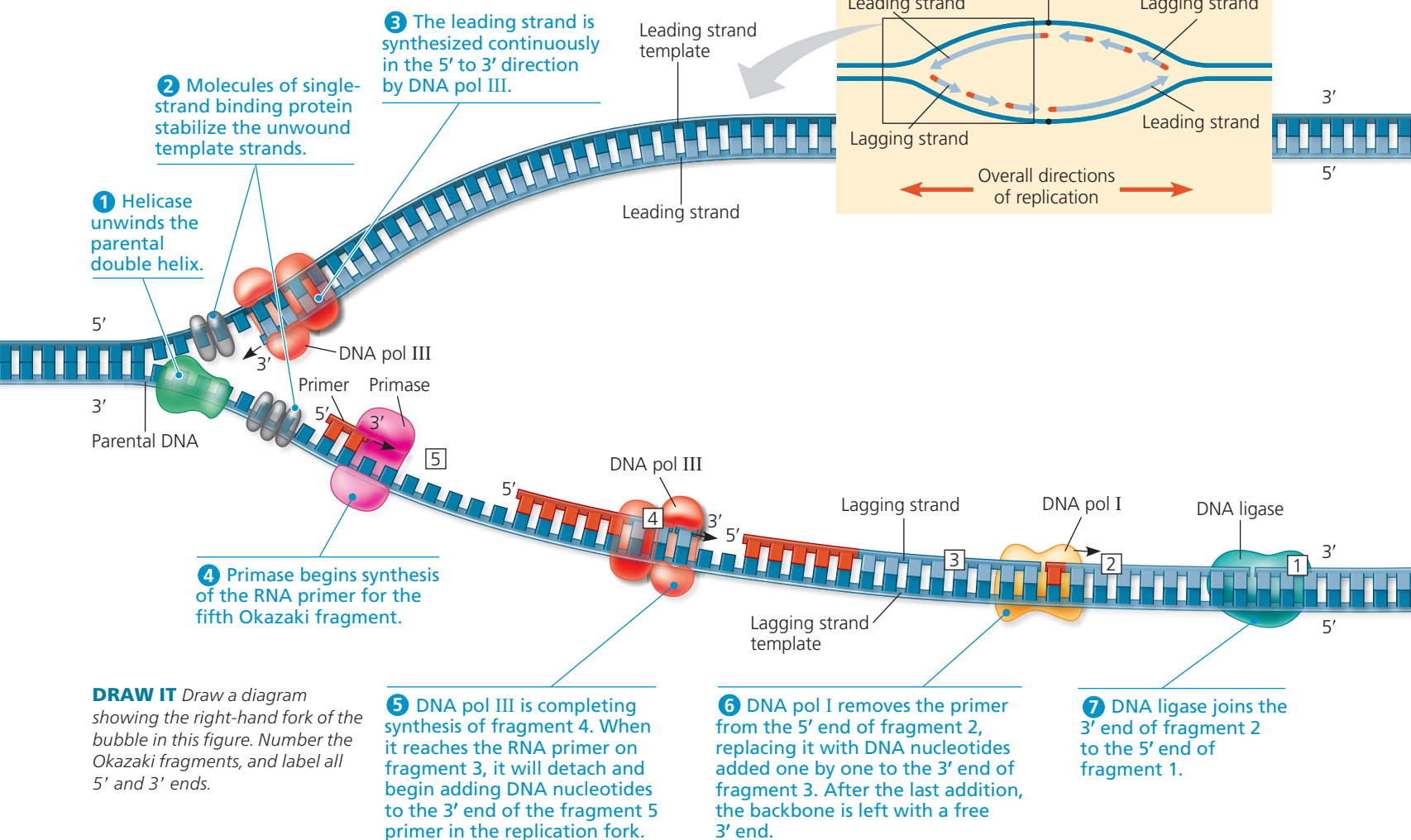
The DNA Replication Complex

It is traditional—and convenient—to represent DNA polymerase molecules as locomotives moving along a DNA railroad track, but such a model is inaccurate in two important ways. First, the various proteins that participate in DNA replication actually form a single large complex, a “DNA replication machine.” Many protein-protein interactions facilitate the efficiency of this complex. For example, by interacting with other proteins at the fork, primase apparently acts as a

molecular brake, slowing progress of the replication fork and coordinating the placement of primers and the rates of replication on the leading and lagging strands. Second, the DNA replication complex may not move along the DNA; rather, the DNA may move through the complex during the replication process. In eukaryotic cells, multiple copies of the complex, perhaps grouped into “factories,” may be anchored to the nuclear matrix, a framework of fibers extending through the interior of the nucleus.

Some experimental evidence supports a model in which two DNA polymerase molecules, one on each template strand, “reel in” the parental DNA and extrude newly made daughter DNA molecules. In this so-called *trombone model*, the lagging strand is also looped back through the complex (**Figure 16.19**). Whether the complex moves along the DNA or whether the DNA moves through the complex, either anchored or not, are still open, unresolved questions that are under active investigation.

▼ **Figure 16.18 A summary of bacterial DNA replication.** The detailed diagram shows the left-hand replication fork of the replication bubble shown in the overview (upper right). Viewing each daughter strand in its entirety in the overview, you can see that half of it is made continuously as the leading strand, while the other half (on the other side of the origin) is synthesized in fragments as the lagging strand.



DRAW IT Draw a diagram showing the right-hand fork of the bubble in this figure. Number the Okazaki fragments, and label all 5' and 3' ends.

➔ **Mastering Biology Animation: DNA Replication: A Closer Look**